



Work, Energy and Power · Worksheet 1

Further Mechanics 1 Chapter 2: Work, Energy and Power. Answer all questions. Show your working clearly. Total: 66 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ ms}^{-2}$. Give work and energy in J, power in W or kW, and non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Work done and kinetic energy	19	
B Potential energy and the work–energy principle	22	
C Power	20	
Roar-level challenge	5	
Total	66	

Section A Work done and kinetic energy

19 marks

1. A box is pulled a distance of 5 m along a horizontal floor by a constant force of 20 N acting at 30° above the horizontal. [2]
Find the work done by the force.

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2. The speed of a particle of mass 4 kg increases from 3 ms^{-1} to 7 ms^{-1} . [2]
Find the increase in the kinetic energy of the particle.

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3. A particle of mass 2 kg is pulled a distance of 10 m along rough horizontal ground by a horizontal force of 12 N. The coefficient of friction between the particle and the ground is 0.25. [3]
Find the work done by the 12 N force and the work done against friction.

4. A crate of mass 25 kg is pulled at constant speed a distance of 8 m up a line of greatest slope of a rough plane inclined at 20° to the horizontal. The coefficient of friction between the crate and the plane is 0.3. [4]
Find the work done against gravity and the work done against friction.



5. A particle of mass 3 kg is pulled from rest along rough horizontal ground by a constant force of magnitude 30 N, which acts at an angle of 25° above the horizontal. The coefficient of friction between the particle and the ground is 0.4. The particle moves a distance of 6 m.

(a) Find the work done by the 30 N force. [2]

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(b) Find the work done against friction. [3]

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(c) Use the work–energy principle to find the speed of the particle after it has moved 6 m. [3]

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(Total for Question 5 is 8 marks)

**Section B Potential energy and the work–energy principle**

22 marks

6. A ball of mass 0.5 kg is lifted vertically through a height of 12 m . [2]
Find the gain in gravitational potential energy of the ball.

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7. A particle is projected vertically upwards from the ground with speed 14 ms^{-1} . Air resistance is [3]
ignored.
Use conservation of energy to find the greatest height reached, and the speed of the particle when it
is 5 m above the ground.

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8. A bead slides along a smooth curved wire. It passes a point A with speed 3 ms^{-1} and later passes a [3]
point B , which is 2.5 m vertically below A .
Find the speed of the bead at B , and explain why *suvat* equations could not be used here.

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9. A particle of mass 2 kg is released from rest and slides 5 m down a line of greatest slope of a rough [4]
plane inclined at 30° to the horizontal. The coefficient of friction between the particle and the plane
is 0.2 .
Use the work–energy principle to find the speed of the particle after it has moved 5 m .

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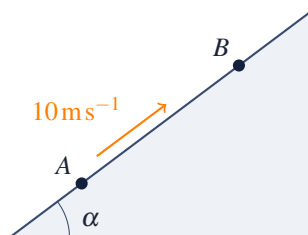
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10. A particle P of mass 4 kg is projected with speed 10 m s^{-1} from a point A , up a line of greatest slope of a rough plane inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$. The coefficient of friction between P and the plane is $\frac{1}{4}$. P comes to instantaneous rest at the point B .



- (a) Use the work–energy principle to find the distance AB .

[5]

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- (b) Determine whether P remains at rest at B .

[2]

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- (c) Find the speed of P when it returns to A .

[3]

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(Total for Question 10 is 10 marks)



Section C Power

20 marks

11. The engine of a car is working at a rate of 12 kW as the car travels at a constant speed of 20 m s^{-1} along a straight horizontal road. [2]
Find the driving force of the engine.

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12. A car of mass 1000 kg moves along a straight horizontal road against a constant resistance of 800 N. The engine of the car is working at 24 kW. [3]
Find the acceleration of the car at the instant when its speed is 15 m s^{-1} .

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13. A van of mass 900 kg has a maximum speed of 40 m s^{-1} on a straight horizontal road, against a constant resistance of 600 N. [3]
Find the maximum power of the engine, and the acceleration of the van when it is working at this power and moving at 20 m s^{-1} .

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14. A lorry of mass 5000 kg moves up a straight road inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{20}$. The resistance to motion from non-gravitational forces is constant at 2000 N. The maximum power of the engine is 60 kW. [4]
Find the maximum speed of the lorry up the road.

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15. A car of mass 1200 kg moves along a straight road. The resistance to motion from non-gravitational forces has constant magnitude R N. When the engine is working at 36 kW, the maximum speed of the car on a horizontal road is 40 ms^{-1} .

(a) Find the value of R . [2]

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(b) The car now moves up a hill inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{14}$, with the engine working at 36 kW. Find the maximum speed of the car up the hill. [3]

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(c) Find the acceleration of the car up the hill at the instant when its speed is 12 ms^{-1} , with the engine still working at 36 kW. [3]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A car of mass 800 kg moves along a straight road against a resistance of magnitude kv N, where $v \text{ ms}^{-1}$ is its speed and k is a constant. With the engine working at 18 kW, the maximum speed of the car on a horizontal road is 30 ms^{-1} . [5]
Find k , and find the maximum speed of the car, with the engine working at 18 kW, up a hill inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{16}$.

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END OF WORKSHEET

Total: 66 marks